4.2.34 V-Sync

Vertical Synchronisation refers to the limiting of the graphics card's rendering rate to match the vertical refresh rate of the monitor.

4.2.35 Vertex

Any point in a 3D space is called a vertex, and can be defined by its coordinates on the X, Y, and Z axes.

4.2.36 VIVO

Video In / Video Out refers to the ability of a graphics card accept various video inputs like Component , Composite, or S-video, and also send out its signals to external devices through these connections. This feature is available in graphics cards that also provide video capture functionality through additional hardware. Incoming video can be recorded by the graphics card and stored to the hard disk, while the video-out function allows signals to be displayed on larger screens like a TV set, which is preferable when watching videos. The GPU does not play a role in video capture.

4.2.37 Workstation Graphics Cards

This term refers to those cards that are specially tailored for use in Workstations—computers used for pursuits like Computer Aided Design and Video Editing. Internally, they are based on the same architecture as their Desktop counterparts. ATI and NVIDIA have workstation graphics cards under the brand names FireGL and Quadro respectively.

4.2.38 Z Buffer

In a 3D scene, objects that are close to the viewer block objects behind them. From a computational point of view, both objects have the same X and Y coordinates of the screen, but differ in the Z coordinate, which represents depth. A Z buffer is the storage area where objects that have a greater Z axis value (and hence are not in view) are stored. This makes it easier to reproduce them when a perspective change requires that object to be displayed.